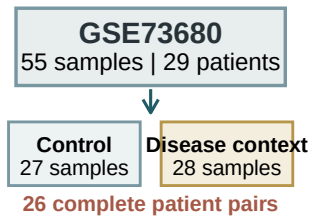


Paired bulk disease context and tissue-state sensitivity

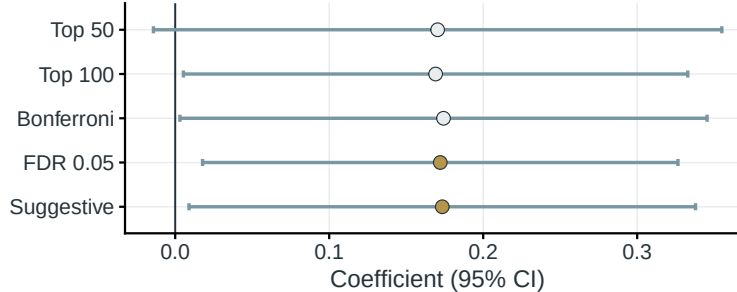
GSE73680: 55 samples from 29 patients, including 26 complete pairs

A paired bulk design



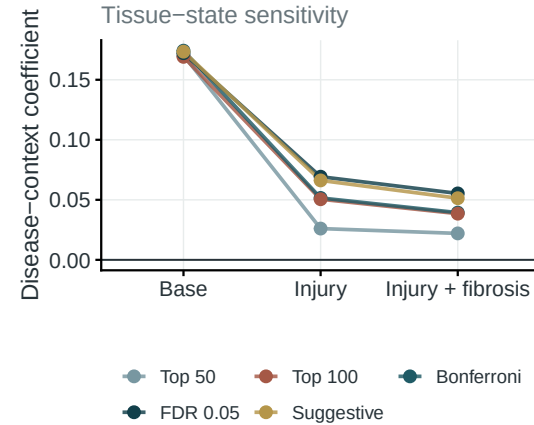
C base disease-context coefficients

All module FDR values > 0.05

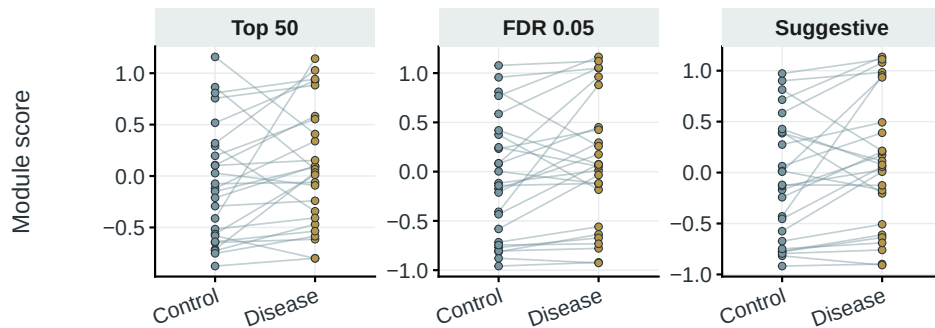


F coefficient attenuation

Tissue-state sensitivity

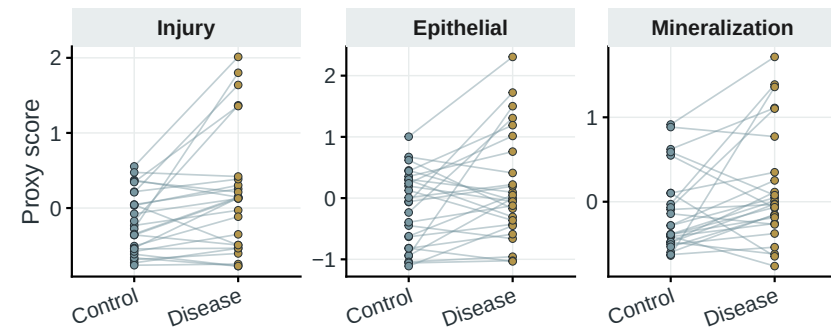


B selected paired module scores



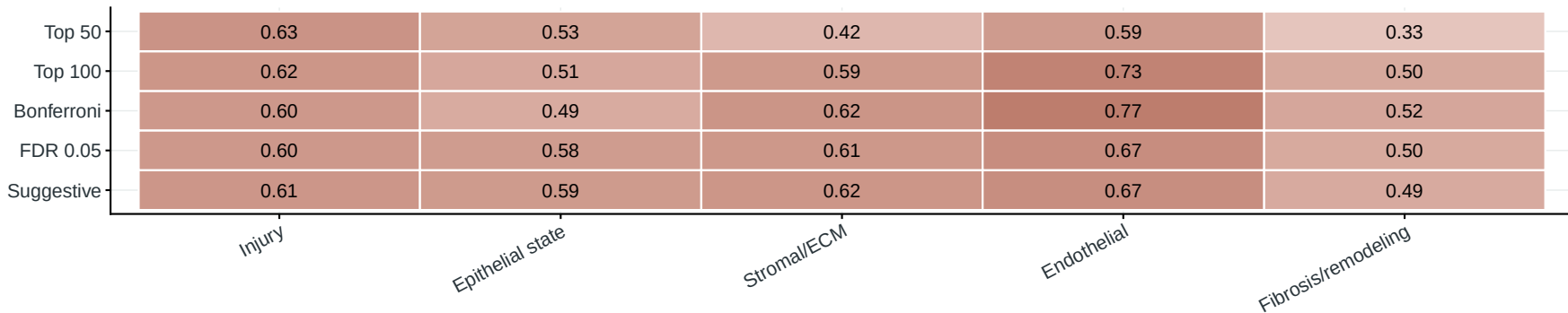
D paired tissue-state proxy shifts

Expression scores, not cell fractions



E module-tissue-state coupling

Paired disease-minus-control differences



Positive module coefficients did not remain FDR significant and attenuated after tissue-state adjustment.